

COURSE INTRODUCTION

Mastering MongoDB

Document modeling, queries, aggregation and production operations — three days on one training_store database.

Day 1
Run MongoDB and model documents

Day 2
Query and transform data

Day 3
Tune, scale and ship

WELCOME TO THE COURSE

A warm welcome, a few ground rules and how we'll work together.



WELCOME

- We're glad you're here
- Three days of learning, building and sharing ideas
- Practical MongoDB skills you can use right away



IN THIS COURSE, YOU WILL

- Work hands-on in mongosh from the first morning
- Use one realistic dataset, training_store
- Compare answers and ideas with your peers
- Get guidance and feedback on your lab work



GROUND RULES

BE OPEN

Ask questions early — even the ones that feel basic.

BE RESPECTFUL

Every experience level in the room adds something.

BE ENGAGED

Run every lab step yourself and check the expected result.

KEY TAKEAWAY Three days of hands-on learning — ask early, try everything, share ideas.

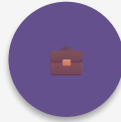
LET'S GET TO KNOW EACH OTHER

Take a minute to introduce yourself to the group.



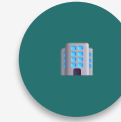
Your name

What should we call you?



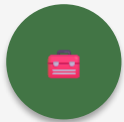
Your role

What do you do day to day?



Your team

Which team or project are you joining from?



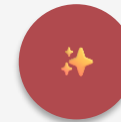
Your experience

SQL, NoSQL, JSON — or a language with a driver?



Your goal

What do you hope to take back to your team?



Something about you

A hobby, a fun fact or a favourite place.

HOW IT WORKS About one minute each. The instructor goes first.

KEY TAKEAWAY Knowing the room helps us connect MongoDB to your real work.

COURSE GOALS AND EXPECTED OUTCOMES

What's in the course — and what you'll be able to do with it.

3

days

~6 hours of teaching a day

8

modules

From NoSQL to production

7

day labs

Two, two, then three

4

collections

One training_store database

1

sample app

Node.js or Python driver



YOU WILL LEARN TO

- Explain NoSQL design and where MongoDB fits
- Design schemas around access patterns
- Create, read, update and delete documents
- Transform and analyze data with aggregation
- Apply indexes that speed up real queries



AND TO RUN IT WELL

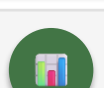
- Describe how replication and sharding help
- Connect MongoDB to application code
- Diagnose common query and operations issues
- Judge whether a deployment is production-ready

KEY TAKEAWAY Eight modules and seven labs take you from document thinking to production-ready operations.

WHO THIS COURSE IS FOR

The audience, the prerequisites and the format.

WHO THIS COURSE IS FOR

-  **Software engineers** Build features on document data
-  **DevOps engineers** Run, secure and monitor deployments
-  **Data engineers** Move and reshape data with pipelines
-  **Data scientists** Query and aggregate data for analysis
-  **Document modelers** Anyone modeling NoSQL documents

PREREQUISITES

-  No prior MongoDB experience
-  Familiarity with any programming language
-  Basic command-line use

COURSE FORMAT

Instructor-led

Virtual or classroom

**Introductory to
intermediate**

Local or cloud MongoDB

NEW TO NOSQL? Module 1 starts from the beginning: why document databases exist at all.

KEY TAKEAWAY No MongoDB experience needed — any programming language and a command line are enough.

THE THREE-DAY ROADMAP

Foundations, then daily data work, then performance and production.



THREE DAYS, ONE DATABASE

Day 1
Run MongoDB and model documents

Day 2
Query and transform data

Day 3
Tune, scale and ship

MODULES 1–3

Why NoSQL exists, a working instance for everyone, then document design.

Lab 1 Install, connect and verify

Lab 2 Model and populate

MODULES 4–5

Find, filter and change documents — then reshape them in pipelines.

Lab 3 Complex queries and updates

Lab 4 Aggregation pipeline

MODULES 6–8

Make Day 2's queries fast, keep data available, and get production-ready.

Lab 5 Index and explain

Lab 6 Replication and sharding

Lab 7 Production readiness

KEY TAKEAWAY Each day ends by applying its modules in hands-on labs on the same data.

THE EIGHT MODULES AT A GLANCE

Grouped by the day we teach them.

DAY 1

Run MongoDB and model documents

1

Introduction to NoSQL Databases

Why NoSQL, four models, documents, architecture and fit

2

Installation and Setup

Deploy, connect with mongosh and Compass, then verify

3

Data Modeling with MongoDB

Access patterns, embed or reference, patterns, validation

DAY 2

Query and transform data

4

The MongoDB Query Language

Filter, project, insert, update, upsert and delete

5

The Aggregation Framework

Totals, rankings and reports — one stage at a time

DAY 3

Tune, scale and ship

6

Indexing and Query Performance

COLLSCAN vs IXSCAN, ESR, explain() and covered queries

7

Introduction to Replication and Sharding

Replica sets, failover, read and write guarantees, shard keys

8

Best Practices, Security, and Troubleshooting

Model, tune, secure, recover and diagnose

KEY TAKEAWAY Aggregation sits next to the query language, and indexing opens Day 3 by tuning it.

ONE DATASET THROUGH THE COURSE

training_store: a small online store you model, query, tune and ship.

CUSTOMERS · 6

Nested name and contact, a bounded addresses array

PRODUCTS · 13

LAPTOP, SHOE, BOOK and ACCESSORY, each with its own attributes

ORDERS · 17

13 PAID. Embeds line items; references the customer

REVIEWS · 6

Ratings 3–5, stored apart from the product



ONE DATABASE, EVERY DAY

Day 1
Model, build and load

Day 2
Query, update and aggregate

Day 3
Index, replicate and ship



FRESH START ON DAY 2 AND DAY 3

```
# PowerShell, from the course repository root
mongosh "mongodb://localhost:27017" .\datasets\training_store\load.js
```

KEY TAKEAWAY The same customers, products, orders and reviews appear in every module and lab.

HOW EACH MODULE RUNS

Short concept blocks, practice and checks — then build it in the lab.



THE RHYTHM OF EACH MODULE



THEN THE DAY LAB

Each day closes with sequenced labs on training_store: two, two, then three.

NEVER STUCK

Every step has a Do this and an Expected result. Reload load.js to catch up.

ASK ANY TIME

Questions in the room or in chat are always welcome — lab time is for trying things.

LOW STAKES Knowledge checks are for you — answer in your own words, then compare with a partner.

KEY TAKEAWAY Learn a concept, see it in mongosh, practise it, check it, then use it in a lab.

SEVEN HANDS-ON DAY LABS

Sequenced labs on training_store, each with Do this and Expected result steps.

DAY 1	Lab 1 — Install, connect and verify	Ping, then insert, find, update, delete	40 min
DAY 1	Lab 2 — Model and populate	Load training_store; justify embed vs reference	50–90 min
DAY 2	Lab 3 — Complex queries and updates	Filters, projections, arrays, verified updates	60–75 min
DAY 2	Lab 4 — Aggregation pipeline	Rank top-selling categories for completed sales	45–60 min
DAY 3	Lab 5 — Index and explain	Baseline, compound and multikey indexes	40–50 min
DAY 3	Lab 6 — Replication and sharding	Pick the mechanism, score shard keys	30–40 min
DAY 3	Lab 7 — Production readiness	Fix, validate, restore, then score go-live	45–60 min

TIME-BOXED TRACK Labs 1–5 and 7. Lab 6 needs a replica-set URI — or do its discussion steps.

KEY TAKEAWAY Every lab runs on live training_store data — reload it and you can always catch up.

LAB ENVIRONMENT AND TOOLS

Where your MongoDB runs, the tools you use, and how to keep secrets safe.



ONE PATH EACH — THE INSTRUCTOR ASSIGNS IT



Local Community Edition

mongod on your own machine at
mongodb://localhost:27017



Managed cloud (Atlas)

A database user, your IP allowed
and a mongodb+srv:// URI



Instructor URI

A shared server — the URI is sent
securely, never in chat



YOUR TOOLKIT

mongosh

MongoDB Compass

PowerShell

Cursor or VS Code

training_store load.js

sample-app: Node or Python



PASSWORD HYGIENE



Let mongosh prompt — no password in the URI



Use placeholders such as <username> in notes



Keep the URI in the MONGODB_URI variable



Never paste passwords in chat or screenshots

KEY TAKEAWAY Every path uses the same mongosh steps — and no password ever lands in chat.

WHAT YOU'LL BE ABLE TO DO BY DAY 3

Seven skills, each proved in a lab.

- ✓ Connect to MongoDB and prove the environment works **Lab 1 · Day 1**
- ✓ Explain the document model — and when to embed or reference **Lab 2 · Day 1**
- ✓ Write filters, projections and verified updates **Lab 3 · Day 2**
- ✓ Build a multi-stage aggregation pipeline **Lab 4 · Day 2**
- ✓ Read an explain() plan: COLLSCAN or IXSCAN **Lab 5 · Day 3**
- ✓ Choose replication, sharding or both for a requirement **Lab 6 · Day 3**
- ✓ Say whether training_store is production-ready — and why **Lab 7 · Day 3**

THE SUCCESS BAR Do all seven by the end of Day 3. (It isn't production-ready yet — you'll name why.)

KEY TAKEAWAY You leave able to model, query, aggregate, tune, scale and assess MongoDB.

LET'S GET STARTED

Questions Before We Begin?

Next: Module 1 — Introduction to NoSQL Databases. Then Module 2 sets up MongoDB for Lab 1.